SS2 PHYSICS LESSON NOTES (THIRD TERM)
WEEK 7

1. INTRODUCTION

In classical physics, matter and energy were treated as distinct entities:

Matter exhibited particle-like properties
Energy (e.g., light) displayed wave-like behavior

However, experiments in the 20th century revealed that both light and matter can exhibit **both wave-like and particle-like properties** depending on the context. This fundamental concept, known as **wave-particle duality**, forms the bedrock of modern quantum physics.

2. CONCEPT OF WAVE-PARTICLE DUALITY AND HISTORICAL CONTEXT

Wave-particle duality refers to the quantum mechanical concept that every particle or quantum entity can exhibit both wave and particle characteristics.

This idea was first introduced in the early 20th century through two critical observations:

Observation	Year	Scientist	Contribution
Photoelectric Effect	1905	Albert Einstein	Explained how light (traditionally considered a wave) could behave like a stream of particles (photons) when it knocks electrons off a metal surface — demonstrating the **particle nature of light**
Electron Diffraction	1927	Davisson & Germer	Electrons (usually treated as particles) produced interference patterns when passed through a crystal lattice or double-slit apparatus — behaviors typical of **waves**, demonstrating the **wave nature of matter**

These discoveries suggested that both light and particles of matter (like electrons, protons, etc.) can behave as both waves and particles, challenging the classical distinction between the two.

3. DE BROGLIE'S HYPOTHESIS AND DE BROGLIE WAVELENGTH

In **1924**, **Louis de Broglie**, a French physicist, extended the wave-particle duality of light to material particles such as electrons, protons, neutrons, and even larger objects theoretically. He proposed that **every moving particle has a wave associated with it**, called the **matter wave**.

De Broglie's Equation:

$$\lambda = h / (mv)$$

Where:

Symbol	Quantity	Unit
λ	Wavelength of the matter wave	meters (m)
h	Planck's constant	6.626×10^{-34} Js
m	Mass of the particle	kilograms (kg)
v	Velocity of the particle	meters per second (m/s)

Key Implications:

Lighter particles moving at higher velocities exhibit **more pronounced wave-like behavior**

For **macroscopic objects**, the wavelength is so small it is undetectable

For **microscopic particles** like electrons, the wave nature becomes significant and observable

4. EXPERIMENTAL EVIDENCE FOR WAVE NATURE OF PARTICLES

One of the most compelling pieces of evidence for the wave-like nature of particles came from **electron diffraction experiments**.

4.1 Davisson-Germer Experiment (1927)

Aspect	Details
Scientists	Clinton Davisson and Lester Germer
Procedure	Electrons were directed at a nickel crystal
Observation	The scattered electrons formed a **diffraction pattern** on a detector — something only waves do
Conclusion	Confirmed that electrons exhibit wave-like behavior, validating de Broglie's hypothesis

4.2 Electron Double-Slit Experiment

Aspect	Details
Procedure	Electrons are passed through a double slit **one at a time**

Aspect	Details
Observation	They still produce an **interference pattern** on the screen
Conclusion	Shows that even individual particles exhibit **self-interference**, a hallmark of wave behavior

These experiments demonstrate that particles such as electrons, under suitable conditions, behave not just like particles but also like waves.

5. IMPLICATIONS OF WAVE-PARTICLE DUALITY

Implication	Explanation
Limits of Classical Physics	The duality showed that classical physics couldn't fully describe the behavior of subatomic particles. It led to the development of **quantum mechanics**, a new framework to explain such phenomena.
Probability in Quantum Physics	Since particles behave like waves, their positions and momenta cannot be precisely predicted. Instead, they are described in terms of **probability waves** (wave functions).
Uncertainty	Werner Heisenberg's uncertainty principle emerges naturally from wave-particle duality.

Implication	Explanation
Principle	The more precisely we know a particle's momentum, the less precisely we can know its position, and vice versa.
Atomic Structure	The wave nature of electrons explains why electrons exist in discrete energy levels around the nucleus and do not spiral into it, as classical physics would suggest.

6. TECHNOLOGIES AND APPLICATIONS BASED ON WAVE-PARTICLE DUALITY

Technology	Description	Principle
Electron Microscope	Utilizes the wave properties of electrons to resolve structures much smaller than what light microscopes can detect. Since electrons have much smaller wavelengths than visible light, they offer higher resolution.	Wave nature of electrons
Scanning Tunneling	Relies on quantum tunneling of electrons, a direct result of wave-	Quantum

Technology	Description	Principle
Microscope (STM)	like behavior, to image surfaces at the atomic level.	tunneling
Quantum Mechanics & Quantum Computing	The entire field of quantum physics is built upon the principles of wave-particle duality. Quantum computers leverage the dual nature of particles (qubits) to perform complex computations more efficiently than classical computers.	Superposition and duality
Particle Accelerators	Use wave-particle properties to accelerate subatomic particles to high energies and study their interactions.	Wave properties of particles
Semiconductor Physics	The behavior of electrons in semiconductors, which power all modern electronics, is described using wave mechanics.	Wave mechanics

7. SUMMARY OF KEY FORMULAS

Formula	Name	Variables

Formula	Name	Variables
$\lambda = h / (mv)$	de Broglie wavelength	λ = wavelength, h = Planck's constant, m = mass, v = velocity

Constants

$h = 6.626 \times 10^{-34}$ Js (Planck's constant)

8. EVALUATION (WAEC/NECO PAST QUESTIONS)

Question 1 (WAEC June 2015 Q2)

(a) Define wave-particle duality.

(b) State two experimental observations supporting the wave nature of electrons.

Question 2 (WAEC June 2022 Q1)

A particle of mass 9.1×10^{-31} kg moves with a speed of 2.0×10^{6} m/s. Calculate its de Broglie wavelength. (Take Planck's constant $h = 6.63 \times 10^{-34}$ Js)

Question 3 (NECO 2020 Q3)

Explain why the wave nature of a moving car is not observable, but that of an electron is.

Question 4 (WAEC June 2017 Q4)

Describe the significance of de Broglie's hypothesis in the development of modern atomic theory

9. ASSIGNMENT

Derive the de Broglie equation from the relationship between momentum and wavelength.
State and explain two technological devices that utilize the wave nature of particles
Research and write a short report on how quantum mechanics is applied in modern computing (Quantum Computing).
(WAEC June 2021) A proton has a mass of 1.67×10^{-27} kg and is moving with a velocity of 5.0×10^{5} m/s. Calculate its de Broglie wavelength. (Use $h = 6.63 \times 10^{-34}$ Js)